UNIT 1: Introduction

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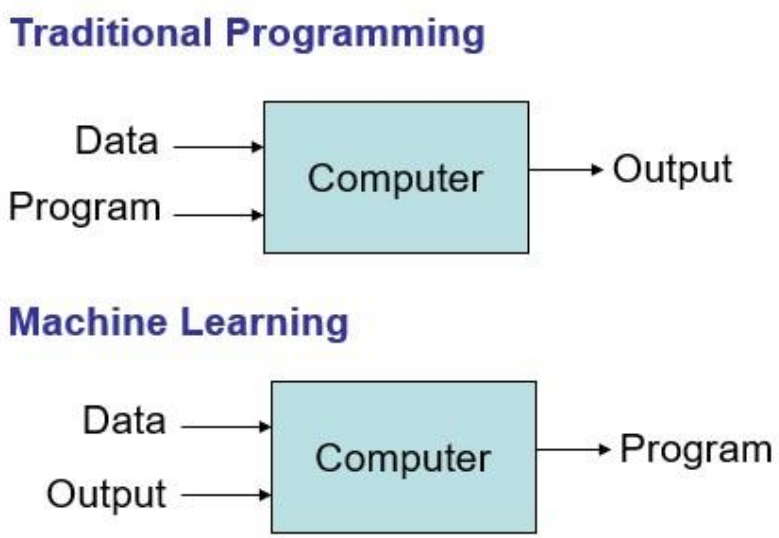
- *(e.g., NumPy, Pandas, scikit-learn)*

What is Machine Learning (ML)?

- ML is the field of study that enables computers to **learn** without being explicitly programmed.
- A computer program is said to **learn** from experience E with respect to some task T and some performance measure P if its performance on T, as measured by P, improves with experience E.
- Machine learning is a subfield of artificial intelligence that involves the development of models that enable computers to **learn** from task (T) and improve their Performance(P) from experience (E).

How machine learning is different from general programming?

- In **general programming**, we have the **data** and the **logic**. By using these two, we create the **answers**.
- But in **machine learning**, we have the **data** and the **answers**, and we let the machine learn the **logic** from them so that the same logic can be used to answer the questions which will be faced in the future



Types of Machine Learning

- Machine learning can be broadly categorized into three main types based on the **learning process** and the **nature of the data** they use. These three types are
  1. Supervised learning
  2. Unsupervised learning
  3. Reinforcement learning
- Each has its own set of algorithms and applications

Types of Machine Learning

Supervised Learning

- **Definition:** In supervised learning, the algorithm is trained on a labelled dataset, where each input example is associated with a corresponding target or output.
- The goal is to learn a mapping from inputs to outputs, enabling the algorithm to make predictions or classifications on new, unseen data.

Examples:

- **Classification:** Predicting whether an email is spam or not spam based on the features of the email.
- **Regression:** Predicting the price of a house based on its features such as square footage, number of bedrooms, and location.

Algorithms: Decision Trees, Random Forest, Support Vector Machines, Logistic Regression, k-nearest Neighbours (k-NN), Linear Regression, etc.

Types of Machine Learning

Unsupervised Learning

- **Definition:** Unsupervised learning involves unlabeled data, where the algorithm seeks to identify patterns, structures, or relationships within the data without a predefined target or output. It is often used for data exploration and dimensionality reduction.

Examples:

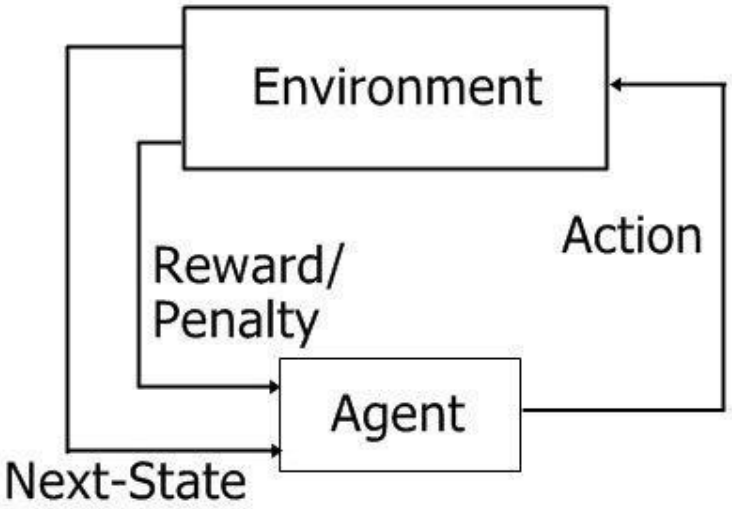
- **Clustering:** Grouping customers into segments based on their purchasing behaviour.
- **Dimensionality Reduction:** Reducing the number of features in a dataset while preserving important information.

Algorithms: K-means clustering, Hierarchical Clustering, Principal Component Analysis (PCA), t-distributed Stochastic Neighbour Embedding (t-SNE), etc.

Types of Machine Learning

Reinforcement Learning (RL)

Definition: RL is focused on training agents to make sequences of decisions in an environment to maximise a cumulative reward signal.



The agent interacts with the environment, receives feedback in the form of rewards or punishments, and learns to take actions that lead to better outcomes over time.

Types of Machine Learning

Reinforcement Learning (RL)

- **RL** is often applied when an agent needs to make sequential decisions, such as in game playing, robotics, and autonomous systems.
  - Game Playing: Teaching a computer program to play chess or Go by learning from its own gameplay and improving over time.
  - Robotics: Training a robot to perform tasks like walking or picking up objects in a real-world environment.
  - Autonomous Vehicles: Reinforcement learning is applied to train autonomous vehicles, such as self-driving cars and drones. RL algorithms enable these vehicles to make informed decisions, navigate through traffic, and learn from real-world driving experiences.
- **Algorithms:** Q-Learning, Deep Q-Networks (DQN), Policy Gradient Methods, Proximal Policy Optimization (PPO), etc.

Types of Machine Learning

- These three main types of machine learning can also be further categorised into other subtypes and specialised techniques.
  - For example, **semi-supervised learning** combines elements of both supervised and unsupervised learning
  - **Transfer learning** involves leveraging knowledge learned from one task to improve performance on another task.
  - Additionally, **deep learning**, a subset of machine learning, focuses on neural networks with many layers and has been particularly successful in tasks involving large amounts of data, such as image and speech recognition.
- The choice of which type of machine learning to use depends on the nature of the problem, the availability of labelled data, and the specific objectives of the task at hand.
- Each type of machine learning has its strengths and weaknesses, making them suitable for different scenarios and applications.

Motivation for Machine Learning

- Machine learning has gained tremendous popularity and attention in recent years due to its ability to enable computers to learn from data and make predictions or decisions without being explicitly programmed.
- There are several key motivations behind the widespread adoption and interest in machine learning:
  - **Automation and Efficiency:** Machine learning algorithms can automate complex tasks and processes that would otherwise require significant manual effort.
  - **Data-Driven Insights:** Machine learning algorithms can analyse and extract valuable insights from this data, helping businesses make informed decisions, identify trends, and discover patterns that might not be obvious through manual analysis.
  - **Continuous Learning and Adaptation:** A key feature of machine learning is its ability to adapt and improve over time.

Applications of Machine Learning

- Machine learning has a wide range of applications across various industries and domains. Here are some notable domains:
  - Image and Video Analysis
  - Natural Language Processing
  - Recommendation System
  - Healthcare
  - Finance
  - Autonomous Systems
  - Environmental Monitoring
  - Entertainment & Gaming
  - Education
- These are just a few domains of the many applications of machine learning. As the field continues to evolve, new applications and innovative uses for machine learning are constantly emerging, contributing to advancements in various domains.

Python and libraries for machine learning

1. NumPy
•Used for **numerical computing**
•Supports **arrays, matrices, linear algebra**
•Faster than Python lists for math operations

2. Pandas
•Used for **data handling and manipulation**
•Provides **DataFrame** and **Series**
•Ideal for CSV, Excel, and tabular data

3. Matplotlib
•Core library for **data visualization**
•Used to create **line plots, bar charts, histograms**

Python and libraries for machine learning

4. Seaborn
- Built on **Matplotlib**
- Provides **statistical and attractive plots**
- Easy visualization with fewer lines of code

5. Scikit-learn
- Library for **Machine Learning**
- Provides tools for:
    - Regression
    - Classification
    - Clustering
    - Model evaluation

How to import them using python

Numerical computation

- import numpy as np

Data manipulation

- import pandas as pd

Basic plotting

- import matplotlib.pyplot as plt

Statistical data visualization

- import seaborn as sns

Machine Learning

- from sklearn.model_selection import train_test_split
- from sklearn.linear_model import LinearRegression
- from sklearn.metrics import mean_squared_error

Google Colab link